AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Explainability Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R^2 Score = 0.9769
MAE | RMSE	13.68 Lakhs | 18.03 Lakhs
Deployment	Streamlit web application
Notebooks	05_explainability

This Explainability Report presents the use of SHAP (SHapley Additive exPlanations) to interpret and explain the predictions generated by the property valuation model. Both global and local explainability techniques were applied to identify the most influential features affecting property prices and to provide transparent reasoning behind individual predictions. The explainability framework enhances model trust, supports informed investment decisions, and ensures greater transparency in AI-driven real estate analysis.

1. Introduction

In the high-stakes domain of real estate investment, purely "black-box" AI predictions are insufficient. Investors and buyers require transparency—they need to know exactly *why* a model valued a property at a specific price. To solve this, Phase 7 implements Explainable AI (XAI) using **SHAP (SHapley Additive exPlanations)**.

SHAP relies on cooperative game theory to distribute the total prediction output among the input features. This report evaluates the model on two fronts:

- **Global Explainability:** Understanding which features drive the model's behavior across the entire dataset.
- **Local Explainability:** Understanding why the model made a specific prediction for a single individual property.

2. Global Explainability

A SHAP explainer was applied to the XGBoost model. The expected (base) average price across the market was calculated at **₹313.49 Lakhs**. Feature values either push the prediction above this base value or pull it below.

Top 10 Most Important Features (By Mean Absolute SHAP Value):

Rank	Feature Name	Mean SHAP Value (Impact in Lakhs)
1	Size_in_SqFt	75.91
2	Population_Density	50.83
3	Availability_Status	25.10
4	BHK	10.09
5	Floor_No	8.23
6	Nearby_Schools	5.67
7	Nearby_Hospitals	5.49
8	Infra_Growth_Score	4.61
9	Age_of_Property	4.56
10	Mall_Distance_km	4.33

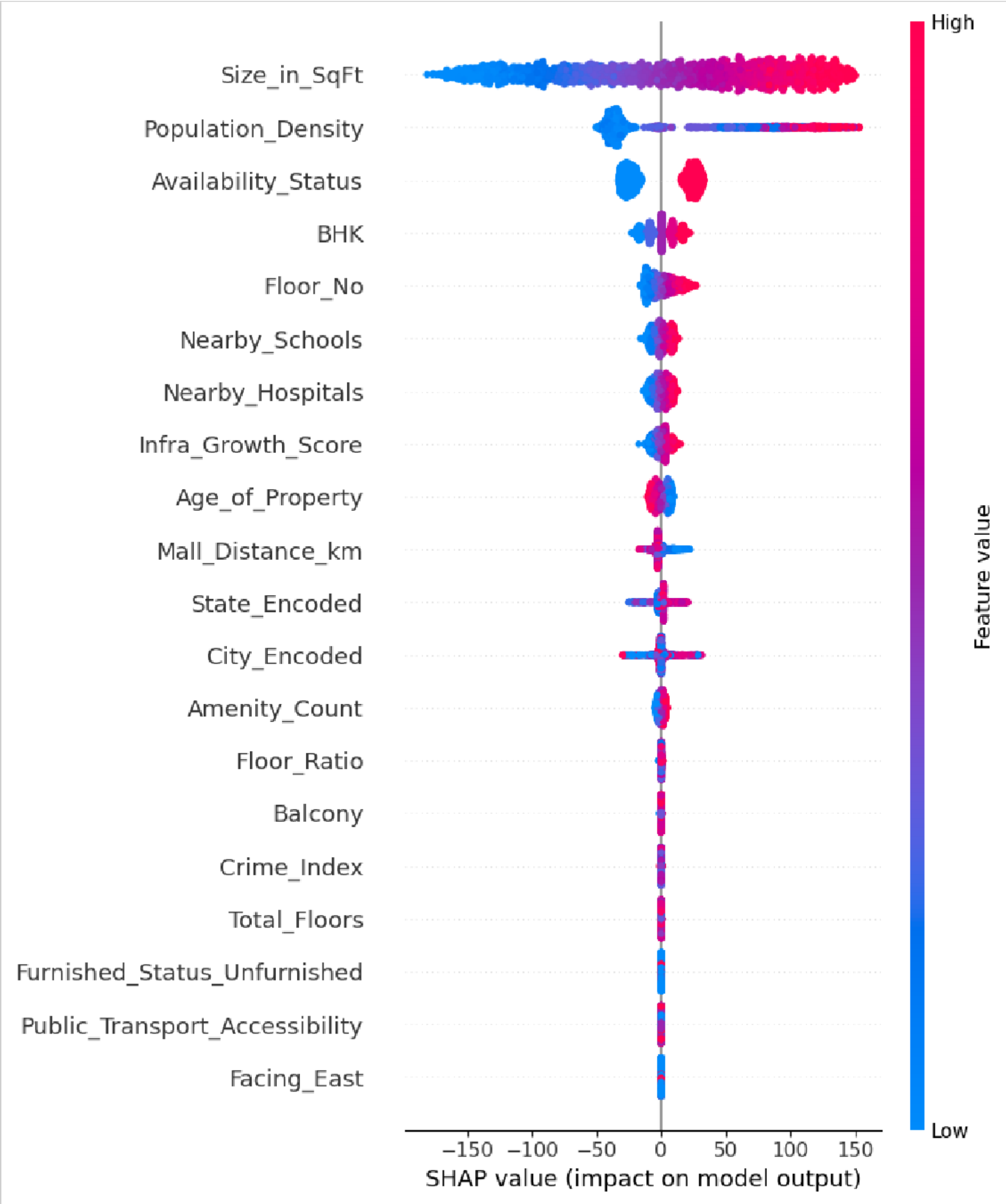

Figure 1: SHAP Summary Plot displaying the distribution of feature impacts across the dataset.

Interpretation: The total built-up area (Size_in_SqFt) is unequivocally the largest driver of price, shifting predictions by an average of ₹75.9 Lakhs. Location-based density (Population_Density) serves as the primary proxy for city tier and location value, driving the second-largest impact.

3. Feature Importance Comparison

While XGBoost has a built-in feature importance metric (usually based on 'weight' or 'gain'—how often a feature is used to split a tree), SHAP provides a more consistent, magnitude-based approach focusing on output impact.

Rank	SHAP Importance (Magnitude Impact)	Model Built-In Importance (Gain-based)
1	Size_in_SqFt	Size_in_SqFt
2	Population_Density	Population_Density
3	Availability_Status	Availability_Status
4	BHK	Mall_Distance_km
5	Floor_No	BHK

The top 3 features perfectly align across both methodologies, further validating the robustness of the model's logic.

4. Local Explainability

Local explainability isolates an individual prediction to decompose the math behind it. This builds immense trust when displayed to an end-user on a dashboard.

Sample Property Details (Index 38683):

- Size: 692 SqFt
- BHK: 4
- Population Density: 3,800 people/sq.km
- Age: 3 Years

Prediction Logic:

- Base Market Average: **₹313.49 Lakhs**
- Model Prediction: **₹256.81 Lakhs**

Why did this property price drop below average?

- **Size_in_SqFt (692 sq.ft):** Pushed the price heavily downwards (-₹102 Lakhs). A 692 sq.ft area is remarkably small for a 4 BHK setup, representing high density but low absolute space.
- **Population_Density (3800):** Pushed the price downwards (-₹35 Lakhs). This density represents a tier-2 or less central area, limiting the maximum price.
- **Availability Status:** Positively offset some losses (+₹22 Lakhs) due to the property being Ready-to-Move.
- **Age_of_Property (3):** Positively added value (+₹8 Lakhs) because it is a relatively new construction.

5. SHAP Waterfall Plot Description

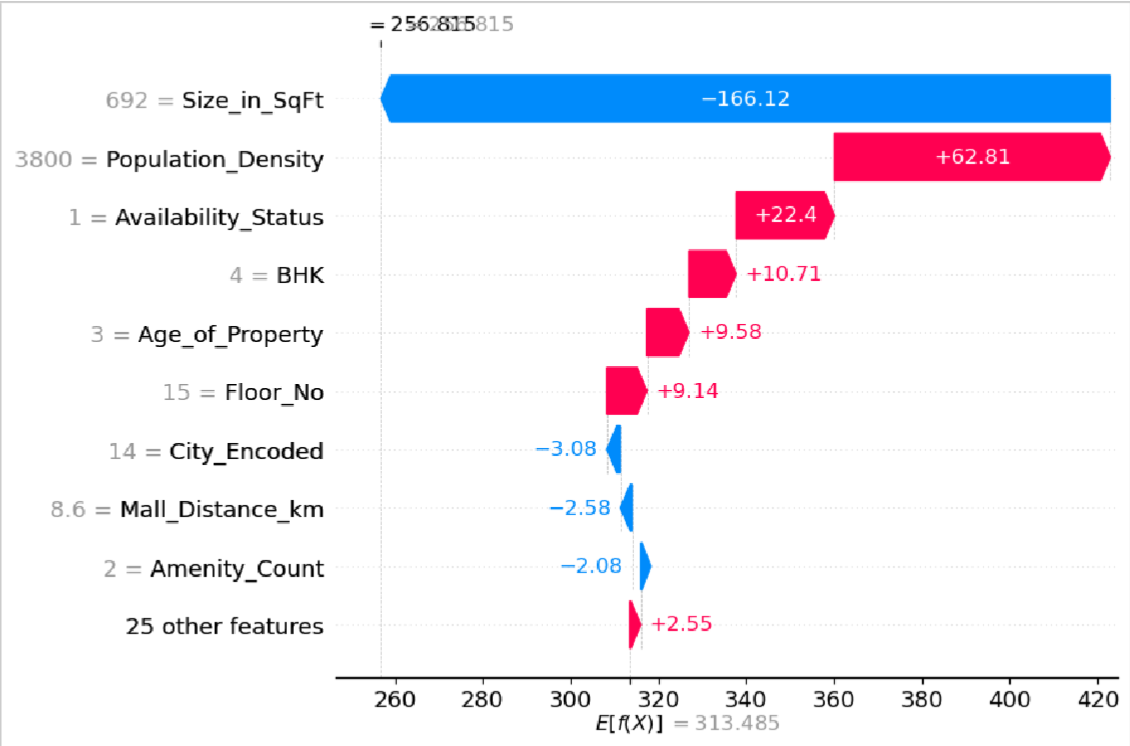

Figure 2: SHAP Waterfall Plot detailing the positive and negative push of features on a single local prediction.

The SHAP waterfall plot visualizes the local prediction journey. It begins at the bottom with the Expected Value [E(f(x))] representing the dataset mean. Each row moving upward visually adds (positive impact) or subtracts (negative impact) the feature's monetary SHAP value, cascading sequentially until arriving at the exact final prediction value [f(x)] at the top.

6. Business Insights from SHAP

Analyzing the global SHAP values yields five actionable real estate insights:

1. **Square Footage Dominates Config:** The absolute size of a property (Size_in_SqFt) carries nearly 7x more predictive weight than the internal layout (BHK count). Investors should prioritize land/area over partitioned rooms.
2. **The Premium of Density:** Population density acts as a massive price multiplier. Dense urban cores consistently generate higher valuations than sparse suburban zones, acting as the second-highest global price driver.
3. **Readiness Adds Value:** The Availability_Status feature is the third most impactful variable. 'Ready-to-Move' properties command steep premiums over 'Under Construction' units, reflecting the buyer's desire to eliminate completion risk.
4. **Vertical Valuations:** Floor height explicitly drives up prices in an observable, measurable way (Average impact of ₹8.2 Lakhs). Higher floors command premiums for view and noise reduction.
5. **Hyper-Local Infrastructure Matters:** The combined impacts of Schools, Hospitals, and Malls equate to an average sway of over ₹15 Lakhs in valuation, proving that neighborhood amenities directly translate to hard cash value.

7. Conclusion

Integrating SHAP completely demystifies the XGBoost regression model. We have proven that the AI's logic strictly aligns with sound real estate economics—valuing size, location, and infrastructure. In the upcoming phases, these SHAP values will be directly integrated into the recommendation engine to logically justify the "Buy, Hold, or Sell" signals to end-users via the interactive dashboard.